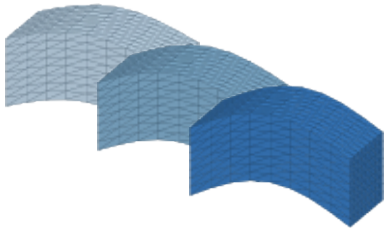
A SCENE ADAPTATION



Observed positions

$\mathcal{D} = \{X_i, x_i(t)\}$

FIT SCENE DESCRIPTORS

$z_m(X),\ z_b$

Geometry, density and correspondences supplied

B SHARED MECHANICS

BULK ENERGY

$W_\theta(F; z_m)$

$P = \partial W / \partial F$

FROZEN SHARED WEIGHTS

ATTACHMENT ENERGY

$\psi_\phi(\delta; z_b)$

FROZEN SHARED WEIGHTS

QUERY / ENVIRONMENT

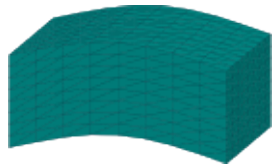
Tool poses / gravity
Contact / initial state

Q2 MECHANICS

$M\ddot{u} = -\nabla_u \Pi$

Compose bulk, attachments, normal contact and body loads

C NEW INTERACTION



Physical state

$x(X,t),\ F,$ reactions

GAUSSIAN TRANSPORT

$\mu_t,\ \Sigma_t \rightarrow$ RGB

Appearance held fixed

CALIBRATION ONLY: position residual → update descriptors

FORECAST: freeze fitted descriptors; change the interaction and roll forward.